

Nhut Hung Nhan

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Summary

Detail-oriented Mathematics student with hands-on experience in computational math, numerical computing, programming, and scientific research.

Experience

Data Research Analyst, *TRG International*, Ho Chi Minh, Vietnam Jul 2026 - Current

- Migrating company's data from MS Projects to Zoho Projects Plus.
- Researching and conducting internal documents such as Standard Operating Procedures (SOP), Business Requirements Document (BRD) and other documents.
- Researching, customizing and documenting business intelligence in order to gain insights from multiple database using Zoho Analytics.
- Replicating, improving and developing reports from SQL Server Reporting Services to Zoho Analytics using SQL.

Teaching Assistant, *Carleton University*, Ottawa, Canada Sep 2023 - Current

- Completed coursework in Numerical Analysis, Ordinary Differential Equations, Mathematical Methods, Mathematical Reasoning, Calculus, and Linear Algebra.
- Recipient of the Mathematics and Statistics Teaching Assistant Excellence Award for 2024–2025.

Teaching Assistant, *Star Education*, Ho Chi Minh, Vietnam Jun 2026 - Aug 2026

- Assisted instruction for Grade 7 and Grade 12 students following the Vietnamese national curriculum.
- Supported students in mathematics problem-solving by clarifying concepts and guiding step-by-step solutions.
- Graded assignments and provided constructive feedback to reinforce learning outcomes.
- Monitored class attendance and maintained accurate student records.
- Collaborated with instructors to ensure smooth classroom operations and effective lesson delivery.
- Delivered individualized tutoring sessions for students to catch up and improve academic performance.

Research Assistant, *Carleton University*, Ottawa, Canada May 2025 - Oct 2025

- Modeling and analyzing quantum wave dynamics in deformed 2D graphene using tight-binding model (Schrodinger base) and finite element approximation model (Dirac base).
- Designing and modifying the model of graphene with lattice deformation for numerical experiments.
- Simulating time-dependent wave propagation using Kwant/tkwant.
- Integrating finite element modeling using FEniCS to solve PDEs and validate deformation models.
- Enhancing results of a time-independent tight-binding solver to compute eigenvalues, band structures, and local density of states (LDOS) for spectral analysis.

Research Assistant, *Carleton University*, Ottawa, Canada May 2024 - Sep 2024

- Utilized Physics-Informed Neural Networks (PINNs) with JAX in Python for solving algebraic fractional systems.
- Conducted numerical experiments in MATLAB, achieving measurable improvements in accuracy or efficiency.
- Enhanced code efficiency for improved computational accuracy and complexity.
- Reviewed and revised research papers.

Skills

Programming:

- Python, Java, SQL, C, C++.
- LaTeX, MATLAB, R, Maple, Julia, Sage, CGSuite.
- HTML, CSS, JavaScript.

Soft Skills: Analytical problem solving, Research rigor, Technical communication, Mentorship and instruction, Collaborative work, Attention to detail, Self-directed learning.

Libraries:

- NumPy, Matplotlib, Pandas.
- JAX, Scikit-learn, PyTorch.
- FEniCS, Pybinding, Kwant.
- MPI.

OS & Tools: Linux, Git, CLI.

Project Management: Microsoft Office, Zoho Projects Plus.

Education

Carleton University Sep 2025 - Current
Master of Science: Mathematics.

Carleton University Sep 2021 - Apr 2025
Bachelor of Mathematics Honours: Concentration in Applied Analysis.

Publications

1. Emmanuel Lorin, Siming Tian and **Howl Nhan**. Simple and accurate microlocal derivation of low energy Dirac operators for locally deformed graphene. *Journal of Physics A: Mathematical and Theoretical*. 59(27):275303, 2026.
2. Emmanuel Lorin and **Howl Nhan**. Data-driven fractional algebraic system solver. *Mathematics and Computers in Simulation*. 236:170–182, 2025.